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Topic: Python Function Parameters and Mutable Defaults

1. Function parameter defaults are evaluated once at definition time.
2. When a default parameter is assigned a mutable container like [] or {},
   subsequent calls share the identical memory reference.
3. To avoid unexpected state retention across calls, use None as default:
   def safe_append(item, target=None):
       if target is None:
           target = []
       target.append(item)
       return target

4. Python binds default arguments to the function object attribute __defaults__.
5. Modifying default arguments in place is one of the most common student bugs.

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Topic: Python Built-in Sequences and Index Boundaries

1. Sequences in Python are zero-indexed from 0 to len(seq) - 1.
2. An IndexError is raised whenever an index outside this range is requested.
3. Common off-by-one bug pattern: range(len(items) + 1).
4. Best practice is to iterate directly over elements: for item in items:
5. KeyError is raised when accessing a non-existent key in a dictionary.
6. Always use dict.get(key, default) for safe retrieval with fallbacks.
7. ZeroDivisionError occurs when divisor evaluates to 0 at runtime.